Quant Fundamentals for Interviews

Basic to intermediate — vocabulary, instruments, and the maths you'll be asked about

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How to use this

This is aimed at a **quant developer** interview, not a quant researcher one. The bar is: use the vocabulary precisely, understand what each number means, and know why a wrong label on a risk report is a real incident. You are not expected to derive Black-Scholes.

Where something is India-specific it's flagged, because NSE conventions differ from US ones and getting those right signals that you've actually worked here.

Part 1 — Market Structure

1.1 The order book

A **limit order book** holds resting orders at each price level:

```
       BIDS (buyers)          ASKS (sellers)
  qty   price            price   qty
  500   2449.90          2450.10  300
  800   2449.85          2450.15  600
 1200   2449.80          2450.20  900
```

- **Best bid** — highest price someone will buy at (2449.90). You **hit the bid** to sell.
- **Best ask / offer** — lowest price someone will sell at (2450.10). You **lift the ask** to buy.
- **Spread** = ask − bid = 0.20. The round-trip cost of immediacy.
- **Mid** = (bid + ask)/2 = 2450.00. A reference price — **you cannot trade at it**.
- **Depth** — quantity available at each level. A tight spread on 10 shares is not liquidity.

The single most important practical point: every trade you do crosses the spread. If a calculation uses the mid, it is not a tradable number. In the assessment's arbitrage problem, mid-based edge was 40.23 points and executable edge was 31.21 — **22% of the apparent profit was spread**.

Order types

Type	Meaning
Market	execute now at whatever price. Guaranteed fill, unknown price.
Limit	execute at this price or better. Known price, no guaranteed fill.
IOC (immediate-or-cancel)	fill what you can now, cancel the rest
FOK (fill-or-kill)	fill entirely now or cancel
Stop / stop-loss	becomes a market order when a trigger price is touched
Iceberg	shows only part of the true size

Price-time priority

Orders match by best price first, then by arrival time within a price level. This is why latency matters — at the same price, the earlier order fills first.

1.2 Market making

A market maker quotes **both sides** and profits from the spread, not from direction.

The business: buy at 2449.90, sell at 2450.10, capture 0.20 per round trip. Do it thousands of times a day on tiny margins.

The three risks:

1. **Inventory risk** — you accumulate a position you didn't want. If you keep getting hit on the bid, you're long and the market is falling.
2. **Adverse selection** — the people trading against you know something. If someone lifts your ask aggressively, the price is probably about to rise. You systematically lose to informed flow and profit from uninformed flow.
3. **Latency risk** — the market moves and your quote is stale. Someone picks you off before you can cancel.

Why a market maker watches net delta, not gross volume: they can turn over hundreds of crores in a day and finish flat. Gross volume describes their *activity* (and their fees and capital); **net position** describes the only thing that loses money on a gap — the leftover inventory.

1.3 Indian market specifics

Exchanges: NSE (dominant for equities and F&O), BSE. Clearing via NSCCL / ICCL.

Timings: pre-open 09:00–09:15, continuous 09:15–15:30, closing session after.

Segments: Cash (delivery / intraday), F&O (futures and options), Currency, Commodity (MCX).

Instruments in F&O: - Index futures and options — NIFTY, BANKNIFTY, FINNIFTY - Stock futures and options — on a specified list of ~180 stocks - Index options are **European** and **cash-settled**; stock options are also European now

Lot size — F&O trades in fixed multiples of shares, revised periodically by NSE. **Cash equity has no lot size.** Calling shares "lots" overstates your position by the whole multiple, which is why it's a genuine risk-report error, not just sloppy language.

Circuit limits — typically 20% for most stocks (2%/5%/10% for some), plus market-wide index-based halts at 10/15/20%.

Expiry — index options are weekly and monthly; monthly expiry is the last Thursday. Settlement uses an **averaging window** (last 30 minutes VWAP), specifically so a single print can't be manipulated.

STT (Securities Transaction Tax) — the one that catches people: - Charged on the **sell** side for delivery equity - Charged on **exercise of in-the-money options, on intrinsic value**

That last one is a real trap. A held-to-expiry arbitrage always leaves one leg ITM, and the STT scales with |settlement − strike|, which you don't know when you put the trade on. So a "locked" arbitrage isn't fully locked in India.

SPAN margining — portfolio-based, so it recognises hedges. A conversion (long future, short call, long put) needs far less margin than the three legs gross.

Part 2 — Options

2.1 The basics

A **future** is symmetric: you take upside and downside equally. Payoff is `S_T − F`, a straight line.

An **option** is asymmetric — the buyer has a *right*, the seller an *obligation*.

- **Call**, strike K: the right to **buy** at K. Worth `max(S_T − K, 0)` at expiry.
- **Put**, strike K: the right to **sell** at K. Worth `max(K − S_T, 0)`.

The `max(·, 0)` is the whole subject. It bends the payoff line at K, and everything interesting — volatility sensitivity, gamma, the smile — comes from that single kink.

Buyer: limited loss (the premium), unlimited gain. **Seller:** limited gain (the premium), unlimited loss. That asymmetry is why sellers post margin and buyers don't.

Moneyness

With the index at 48,500:

	Call	Put
ITM (in the money)	K < 48,500	K > 48,500
ATM (at the money)	K ≈ 48,500	K ≈ 48,500
OTM (out of the money)	K > 48,500	K < 48,500

Intrinsic and time value

Intrinsic = what you'd get exercising right now = `max(S − K, 0)` for a call. **Time value** = everything above intrinsic. The market's price for remaining uncertainty.

An ATM option has **zero intrinsic** — it is all time value. Time value is always ≥ 0 (you can always walk away) and decays to exactly zero at expiry.

European vs American

European — exercise only at expiry. All Indian index and stock options. **American** — exercise any time. Adds early-exercise risk for the seller, which matters for conversions and dividend-paying stocks.

2.2 Put-call parity — the one relationship to know cold

Buy a call and sell a put at the **same strike and expiry**. What do you own?

- If `S_T > K`: call pays `S_T − K`, put expires worthless. Payoff `S_T − K`.
- If `S_T < K`: call worthless, the put you sold costs you `K − S_T`. Payoff `S_T − K`.

Same expression both times. The kinks cancel exactly at K, and you're left with a straight line of slope 1. **That is a future.**

Since the payoffs are identical on every path, the prices must be too:

```
C - P = (F - K) * e^(-rT)
```

Rearranged to give the **synthetic futures price**:

```
F_syn = K + (C - P) * e^(+rT)
```

Get the sign of the exponent right. You pay the premium **today**; the futures P&L settles at **expiry**. So the premium difference is carried **forward**, not discounted back. `e^{-rT}` there is the standard slip and it produces an answer that looks perfectly reasonable.

Synthetics

Want	Build from
Long future	long call + short put
Short future	short call + long put
Long call	long future + long put
Long put	short future + long call

Why parity is special

Black-Scholes needs a distribution, constant volatility, continuous trading, no jumps. **Parity needs none of it** — it's just two portfolios with the same payoff. So if parity breaks, *someone is giving away money*; it isn't "the model is wrong."

The structures built on it

- **Conversion** — buy the actual future, sell the synthetic (sell call, buy put). Do it when the synthetic is **rich**.
- **Reversal** — sell the future, buy the synthetic. When the synthetic is **cheap**.
- **Box** — a call spread plus a short put spread at two strikes. Pays exactly $K_2 - K_1$ at expiry whatever happens. A synthetic zero-coupon bond made of options.

2.3 The Greeks

Greek	Definition	What it means
Delta (Δ)	$\partial V/\partial S$	How many shares the option behaves like right now
Gamma (Γ)	$\partial^2 V/\partial S^2$	How fast delta changes — the curvature
Vega (ν)	$\partial V/\partial \sigma$	Sensitivity to implied volatility
Theta (Θ)	$\partial V/\partial t$	Time decay per day
Rho (ρ)	$\partial V/\partial r$	Interest rate sensitivity — usually least interesting

Delta

Runs $0 \to 1$ for a call, $0 \to -1$ for a put. ≈ 0.5 at the money.

Loosely interpretable as "the probability of finishing in the money" — a useful intuition, not exactly true.

For cash equity, delta is exactly 1.0 per share, always. A share gains ₹1 when the stock gains ₹1. There's no optionality. So a 70-share position has a delta of 70 shares, or ₹173,600 of delta-adjusted exposure at a mark of 2480. Saying "our delta is 70%" is a category error.

Delta hedging: hold an offsetting position in the underlying so small price moves don't affect you. It has to be redone as delta changes — and that's gamma.

Gamma — the one to understand properly

Gamma is how fast your delta moves. It peaks **at the money, near expiry**.

Short gamma (what an option seller has): your delta moves *against* you. Price rises → your delta goes more negative → you buy to rehedge, at the higher price. Price falls → you sell, lower. **You systematically buy high and sell low while rehedging.**

The trade-off: you collect **theta** every day, and lose it in one big move. That's the structural risk of a market maker who is net short options.

Long gamma is the mirror: rehedging buys low and sells high, but you pay theta for the privilege.

This is the single most useful Greek intuition to be able to explain out loud.

Vega

Vega is identical for a call and a put at the same strike and expiry. This falls straight out of parity: `C − P` doesn't depend on σ at all, so differentiating gives `∂C/∂σ − ∂P/∂σ = 0`.

That's also why a conversion is safe — short call + long put has **net vega of exactly zero**, so an IV move after the trade is on cannot touch the locked profit.

Vega peaks at the money and dies in the wings. A far OTM option is worth almost nothing and stays worth almost nothing whatever vol does.

Theta

Time decay. Negative if you're long options, positive if short. Accelerates as expiry approaches, fastest for ATM options.

The seller's income statement in one word: **theta is what you collect, gamma is what you pay it for.**

Part 3 — Volatility

3.1 What volatility actually prices

An option's value comes from the kink. If the index were certain to finish at exactly 48,500, both the call and the put at 48,500 would be worth nothing. The more spread out the possible outcomes, the more the `max(·,0)` is worth — **because you keep the good tail and discard the bad one.**

So **an option is a price on dispersion**. More expected movement means a higher price for calls *and* puts alike, regardless of direction.

3.2 The three volatilities

Realised (historical) volatility — the standard deviation of past returns, annualised:

```
sigma_annual = sigma_daily * sqrt(252)
```

(252 trading days. Volatility scales with **√time**, not time, because variance is additive over independent periods.)

Implied volatility — the σ you must feed Black-Scholes to reproduce the market price.

Forecast volatility — what you think will happen. The actual research problem.

IV is a quoting convention, not a prediction. It's an invertible map from price to a number, which strips out strike, expiry and forward so different options become comparable — exactly as a bond *yield* is more useful than a bond *price*.

3.3 The smile and skew

Black-Scholes assumes one σ for all strikes. Reality doesn't cooperate: real returns have fat tails and crashes happen faster than rallies, so OTM puts are bid up relative to what a lognormal would imply.

Plot IV against strike and you get a **smile** (both wings elevated) or a **skew** (puts elevated more). Equity indices show a pronounced downward skew — crash protection is expensive.

The distinction that matters for arbitrage

The smile is about **different strikes** relative to each other. Put-call parity is about the **call and the put at the same strike**.

Whatever shape the smile has, the call and put at one strike must imply the **same** vol — because their price difference is pinned by parity, which doesn't know what vol is.

So a call/put IV gap at one strike isn't smile. It's a parity violation.

In the assessment: 72.5 basis points of gap at one strike, under 1 bp at both neighbours. And repricing those two options off the *synthetic* forward instead of the actual one collapsed the gap to **exactly zero** — proving the IV gap and the forward dislocation are the same fact in different units.

3.4 Black-Scholes — what to know

You don't need to derive it. You need to know what it assumes and where those assumptions fail.

Assumes: lognormal prices, constant volatility, continuous trading with no costs, no jumps, constant risk-free rate, European exercise.

Fails because: volatility isn't constant (that's the smile), returns have fat tails, jumps happen (gaps, results announcements), and trading costs money.

Still used because it's a common language — quoting in "vol" is universal precisely because everyone agrees on the mapping, even though nobody believes the model.

The one structural fact worth knowing: the price is the **discounted risk-neutral expected payoff**. Risk-neutral means we price under the measure where all assets drift at the risk-free rate — not a belief about the real world, just the measure in which no-arbitrage pricing works.

Part 4 — P&L and Exposure Vocabulary

A mislabelled number on a risk blotter is a real incident. This section is mostly about precision of language.

4.1 The terms

Term	Definition
Realised P&L	Banked when a fill *reduces* the position. Permanent, mark-independent.
Unrealised (MTM) P&L	Open position valued at a chosen mark, minus its cost. Re-prices every tick.
Total P&L	realised + unrealised
Gross traded notional	Σ |qty| × price over all fills. **Turnover. Only ever increases.**
Net notional exposure	signed qty × mark. **The directional number.**
Cost basis	What the open position cost — not what it's worth
Delta exposure	position delta × mark

The two most common confusions

Gross traded notional is not exposure. It includes everything you've already closed, and it never goes down. In the assessment's example: gross traded ₹565,600 versus actual net exposure ₹173,600 — an overstatement of 3.3×, widening all day.

Total P&L is not realised P&L. ₹3,200 total = ₹1,333.33 realised + ₹1,866.67 unrealised. The first is banked and cannot change; the second is a mark-dependent estimate that could be negative by the close.

4.2 Cost-basis conventions

How you match a sale against prior purchases changes the **split** between realised and unrealised:

Convention	Realised	Unrealised	Total
Weighted average	₹1,333.33	₹1,866.67	**₹3,200**
FIFO	₹1,600.00	₹1,600.00	**₹3,200**
LIFO	₹1,100.00	₹2,100.00	**₹3,200**

The total is invariant; the split is not. That's the thing to say.

India-specific: capital gains on listed equity are computed **FIFO**. So a weighted-average risk book will *correctly* disagree with the tax book. Reconcile on total, never on the split.

4.3 Choosing a mark price

Unrealised P&L depends entirely on which mark you use, so the choice is a real decision:

Mark	Issue
Last trade	can be stale; manipulable near the close
Mid	not tradable, but unbiased
Bid (for a long) / ask (for a short)	conservative — what you'd actually exit at
Exchange settlement price	authoritative; NSE uses a closing VWAP window
Theoretical / model	needed for illiquid options, but now your P&L depends on your model

Always record which mark produced a number. "Which mark did you use?" is the first question asked about any disputed blotter line.

Part 5 — Execution Quality

5.1 The benchmarks

Slippage is undefined without naming a benchmark. "Our slippage was zero" is not a statement.

Benchmark	Meaning	How it's gamed
Arrival price	price when the decision was made	punishes you for market moves outside your control
Interval VWAP	volume-weighted average over the execution window	rewards trading slowly even as the market runs away
TWAP	time-weighted average	ignores volume entirely
Close	closing price	suits index funds, almost nobody else

Implementation shortfall = the total cost from decision to completion: market impact, timing cost, and the opportunity cost of anything unfilled.

Matching VWAP means you were average, not free. Even a perfect VWAP execution has non-zero shortfall whenever the market moved between decision and fill.

5.2 A worked example

Buy 100 @ 2450 and 50 @ 2460, with an arrival price of 2450.

- Execution VWAP = (100×2450 + 50×2460) / 150 = **₹2,453.33**
- Slippage vs arrival = (2453.33 − 2450) / 2450 = **+13.6 bps adverse**

Sign convention worth adopting: **positive always means worse**, on both sides. A buy above arrival paid up; a sell below arrival sold cheap. Without a convention, half your consumers get the sign backwards.

5.3 Market impact

Trading moves the price against you. Roughly, impact scales with the **square root** of order size relative to daily volume — so doubling your size costs about 1.4× the impact, not 2×.

Temporary impact — the price recovers after you stop. **Permanent impact** — you revealed information; the price stays moved.

This is why large orders are sliced. And why "capacity" is a real constraint on a strategy: a signal that works on ₹1 crore may be entirely eaten by impact at ₹100 crore.

Part 6 — Statistics You'll Be Asked About

6.1 Returns

Simple return: `(P₁ − P₀) / P₀` **Log return:** `ln(P₁ / P₀)`

Log returns are **additive over time** — a 10-day return is the sum of 10 daily log returns, which makes the maths cleaner. Simple returns are additive **across a portfolio**. Use whichever matches the direction you're aggregating.

For small moves they're nearly identical; they diverge for large ones.

6.2 The key measures

Sharpe ratio = (mean return − risk-free) / standard deviation, annualised by ×√252. The standard risk-adjusted return measure. Its flaw: it penalises upside volatility identically to downside.

Sortino — same, but only downside deviation in the denominator.

Maximum drawdown — the largest peak-to-trough decline. Often matters more than Sharpe in practice, because it's what determines whether you survive to keep trading.

Correlation vs covariance — correlation is covariance normalised to [−1, 1], so it's comparable across pairs.
Correlation is not causation, and it isn't stable — correlations famously go to 1 in a crisis, exactly when you were relying on diversification.

Beta — sensitivity to the market: `cov(r_stock, r_market) / var(r_market)`.

6.3 Concepts that come up

Stationarity — statistical properties don't change over time. Prices are non-stationary; returns are closer to stationary. That's why models work on returns.

Mean reversion vs momentum — does a move tend to reverse or continue? Different time horizons often show different answers, which is most of what systematic research is about.

Look-ahead bias — using information that wasn't available at the time. The most common way a backtest lies to you.

Survivorship bias — testing only on companies that still exist today, so you've excluded every failure.

Overfitting — a strategy that works beautifully in-sample and not at all out-of-sample. Guard with out-of-sample testing,

walk-forward validation, and being suspicious of anything with many parameters.

Part 7 — Likely Interview Questions

“What’s the difference between a future and an option?” A future is symmetric — both sides are obliged, and payoff is linear. An option gives the buyer a right and the seller an obligation, so the payoff has a kink at the strike. Everything interesting about options comes from that asymmetry.

“What is delta hedging, and why do you have to keep doing it?” Hold an offsetting position in the underlying so small price moves don’t affect you. You have to redo it because delta itself changes as the price moves — that rate of change is gamma.

“Why does a market maker care about gamma?” Short gamma means rehedging systematically buys high and sells low. You collect theta every day and give it back in one big move. It’s the structural risk of being net short options.

“What does implied volatility mean?” The volatility that reproduces the market price in a model. It’s a quoting convention that makes different strikes and expiries comparable — not a forecast.

“State put-call parity and why it matters.” `C − P = (F − K)e^{−rT}`. It comes from static replication — long call plus short put pays `S − K` on every path — so it holds regardless of any model. If it breaks, someone is giving away money.

“Realised vs unrealised P&L?” Realised is banked when a fill reduces the position; permanent and mark-independent. Unrealised is the open position at a chosen mark; it re-prices every tick and can reverse.

“Notional exposure — define it.” Signed position × mark. Ask which one they mean, though — gross traded notional is turnover and only goes up, net notional is the directional number. Confusing them overstates risk by multiples.

“What’s slippage?” The difference between an execution price and a benchmark. Meaningless until you name the benchmark — arrival price and interval VWAP give different answers and are gamed differently.

“How would you check whether an option is mispriced?” Relative to what? Against parity you can make a model-free statement. Against a smile fit you need a model, and then you’re expressing a view rather than arbitraging.

“Why is volatility annualised with √252?” Variance is additive over independent periods, so standard deviation scales with the square root of time. 252 is the trading-day count.

“What’s the risk in a cash-and-carry arbitrage?” Financing cost, margin calls on the futures leg’s daily mark-to-market, execution risk if you leg in and don’t get all sides, and — in India — STT on exercise, which is charged on intrinsic value and therefore scales with where the index settles, which you don’t know when you enter.

The habit worth carrying in

Precision of language *is* the skill being tested here. “Notional”, “delta”, “slippage” and “realised” all have exact meanings, and a role on a risk-facing desk depends on using them exactly.

If you’re unsure which of two things someone means, **ask** — “gross traded or net?” is a better answer than guessing, and it signals you know they’re different.